
Chapter 1: Patterns in Mathematics

Worksheet 1: Number Sequences and Patterns

Learning Objective: Students will identify and extend number sequences, describe rules for patterns, and explore relationships among number sequences such as counting numbers, squares, and triangular numbers.

Worked Example

Find the next two terms: 1, 4, 9, 16, 25, _____, _____

Solution: These are perfect squares: $1^2, 2^2, 3^2, 4^2, 5^2, \dots$

Next terms: $6^2 = 36$ and $7^2 = 49$.

Questions 1–5: Easy / Recall

Q1. Find the next two terms: 2, 4, 6, 8, 10, _____, _____

Q2. Find the next two terms: 1, 3, 6, 10, 15, _____, _____ (triangular numbers)

Q3. What is the rule for this sequence? 5, 10, 15, 20, 25, ...

Q4. Find the missing term: 2, 4, 8, _____, 32

Q5. Write the first 5 square numbers.

Questions 6–10: Basic Application

Q6. A pattern adds 7 each time. If the first term is 3, write the first 6 terms.

Q7. How many dots are in the 6th triangular number pattern?

Q8. Find the next three terms: 1, 1, 2, 3, 5, 8, _____, _____, _____ (Fibonacci)

Q9. Observe: $1 = 1$; $1 + 3 = 4$; $1 + 3 + 5 = 9$; $1 + 3 + 5 + 7 = 16$. Find $1 + 3 + 5 + 7 + 9 + 11$.

Q10. What comes next? 1, 8, 27, 64, _____ (cube numbers)

Questions 11–15: Practice and Skill Building

Q11. Complete: $1 \times 1 = 1$; $11 \times 11 = 121$; $111 \times 111 = 12321$; $1111 \times 1111 =$ _____

Q12. The 10th triangular number is $\frac{10 \times 11}{2} =$ _____. Find the 12th triangular number.

Q13. A sequence starts at 100 and each term decreases by 11. Write the first 6 terms.

Q14. Find the pattern rule and the 8th term: 3, 6, 12, 24, 48, ...

- Q15.** Show that the sum of any two consecutive triangular numbers is a square number.
(Use $T_3 = 6$ and $T_4 = 10$.)

Questions 16–18: Higher Order Thinking

- Q16.** How many squares (of all sizes) are in a 4×4 grid? (Hint: count 1×1 , 2×2 , 3×3 , 4×4 separately.)
- Q17.** The number of handshakes when n people each shake hands once with every other person is $\frac{n(n-1)}{2}$. How many handshakes occur among 10 people?
- Q18.** A sequence is defined as: $a_1 = 2$, $a_2 = 5$, and each term = sum of previous two terms. Write the first 8 terms.

Questions 19–20: Real-Life Applications

- Q19.** Seats in a cinema hall are arranged so that Row 1 has 10 seats, Row 2 has 13 seats, Row 3 has 16 seats (increasing by 3 each row). How many seats are in Row 8? What is the total number of seats in 8 rows?
- Q20.** In a savings scheme, Priya saves 50 in Week 1, 100 in Week 2, 200 in Week 3 (doubling each week). How much does she save in Week 8? What are her total savings after 8 weeks?

Reflection Question: Can a number be both a triangular number and a square number? Investigate with examples.